ahmad_mahmoud.resume.ts SENIOR SOFTWARE ENGINEER ● 12+ YRS

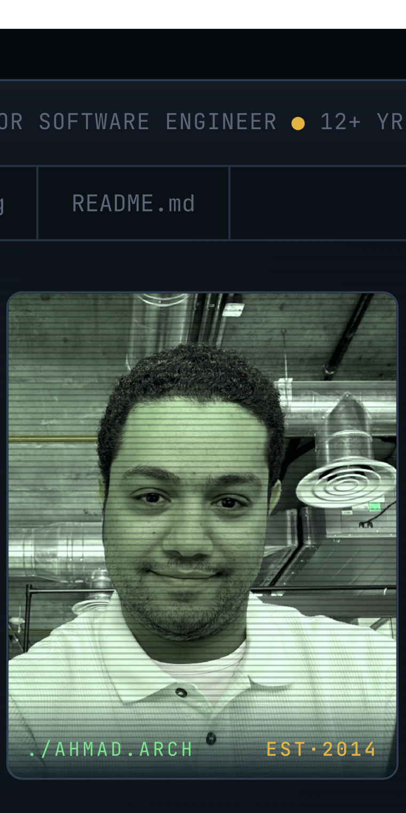
● senior_engineer.ts | experience.json | projects/ | stack.config | README.md

→ whoami --full

Ahmad Abdul-Jawad Mahmoud

Senior Software Engineer / iOS · Android · Backend

// **12+ years** shipping software · **iOS** · **Android** · **backend** · Cairo → Riyadh · apps used by 35M+ people



12+	YEARS HANDS-ON
35M+	USERS REACHED
99.8%+	CRASH-FREE IN PROD
3,700+	TESTS ON MY PLATFORM

```
/**
* Senior Software Engineer — 12+ years shipping production
* software, native iOS first and now across Android, backend and web.
* Apps used by 35M+ people: HungerStation, Ego, and seven years on the
* Marn POS (20k+ restaurants) — an offline-first CoreData core with cloud
* sync at a 99.8-99.9% crash-free rate.
* Right now: a production ride-hailing platform built end to end, solo —
* NestJS / PostgreSQL / Redis, a Next.js console, native iOS & Android.
*/
```

## how_i_engineer()

4 pillars

[01] systems design

Design for the failure case

An offline-first CoreData + cloud-sync core so a venue keeps trading when the network does not. A **trip state machine**, **scored dispatch** over a Redis geo index and a **double-entry ledger** with idempotent mutations, so a retry can never double-charge.

◆ Offline-first · idempotent · double-entry

[02] quality you can measure

Tests gate the merge

Automated tests that block bad merges, CI/CD built from nothing (**60% faster deployments**), and a **99.8-99.9% crash-free rate** in production. On my current platform: **2,900+ backend** and **800+ iOS** tests.

◆ 99.8-99.9% crash-free · 60% faster deploys

[03] ship without stopping

Modernise in flight

A multi-year move to SwiftUI, Combine and async/await without pausing feature delivery, and a performance programme that cut load times **40%**. Swift 6 strict concurrency, actors and structured concurrency in new code.

◆ SwiftUI migration · 40% faster loads

[04] end-to-end ownership

Credible at every layer

Native iOS (SwiftUI) and Android (Kotlin / Compose) clients on shared, versioned core libraries; NestJS / PostgreSQL / Redis backends; a Next.js console. I build and run **all four layers of a production platform myself**.

◆ 4 layers · 3,700+ tests

## experience()

8 roles

iOS Team Lead @ Marn — Jahez Group

Aug 2019 → Aug 2026

RIYADH, KSA

+ **Owned the architecture** of an enterprise POS used by 20k+ restaurants and cafés: an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
+ **Designed the core systems** — table management, order processing and inventory — at the heart of the product.
+ **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery.
+ **Held a 99.8-99.9% crash-free rate** in production through clear module boundaries, code review and automated tests that gate merges.
+ **Built CI/CD and the release process from scratch**, cutting deployment time 60%, and ran the performance programme behind 40% faster load times.
+ Led and mentored a **5+ engineer iOS team** for seven years, and partnered with product, design and backend on scope and estimates.

Founding Platform Architect — iOS · Android · Backend · Web @ Ride-Hailing Platform · Independent

Mar 2026 → Present CURRENT

Took the **full technical scope of a founding CTO** and carried it alone — building a production ride-hailing platform **solo**: a hardened backend and a real-time ops console **in production**, with native **iOS/SwiftUI** and **Android/Compose** apps (Rider & Driver on each) in active development — **owning every layer** and setting the engineering standards (idempotency, double-entry accounting, acyclic module boundaries, CI test suite).

› Backend APIs
// NestJS · TypeScript · Mar '26 → now

Modular monolith, **23 modules** — **trip state machine**, **scored dispatch** over a Redis geo index, **double-entry ledger**, realtime rider/driver/admin **sockets**, JWT + phone-OTP, idempotency & audit log · ~320 endpoints over 47 controllers, traffic-aware fare quotes, **2,900+ tests**, Swagger docs.

› Admin Console
// Next.js 14 · Jun '26 → now

Bespoke **30-screen** ops surface — live **MapLibre** trips map, dispatch/SOS live boards, KYC onboarding review, finance reconciliation, refunds/payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO**.

› iOS Apps · Rider & Driver
// SwiftUI · Jul '26 → now

Two native **SwiftUI** apps on a shared **BuraqCore** Swift package (domain · networking · design system, pinned remotely) — live GPS **trip lifecycle**, Socket.IO feeds with fresh-token reconnect, in-app chat, payments, Live Activities, driver earnings & turn-by-turn navigation · Swift 6 strict concurrency, **800+ tests**.

› Android Apps · Rider & Driver
// Kotlin · Jetpack Compose · Sep '26 → now

Native **Kotlin / Compose** ports of both apps against **the same API contract**, on a shared **buraq-core-android** library published to GitHub Packages — driver KYC onboarding, offers and trips; rider booking, live tracking, FCM push and Google Pay · **290+ test suites**.

One backend · one contract · four native clients in flight — plus growth loops (promotions, referrals, loyalty, wallet), driver safety/SOS, and scheduled reconciliation jobs, owned end-to-end.

Senior iOS Engineer @ HaramBlur · voluntary charity project

Dec 2025 → Jun 2026

+ Designed an architecture integrating **on-device machine learning** for real-time content analysis.
+ Built a seamless communication layer between native iOS and web-based content.
+ Implemented efficient image processing with region-based and full-image blur effects.
+ Created a real-time video monitoring system for continuous content filtering.
+ Developed smart banner blocking and optimised memory management throughout.

iOS Team Lead @ Ego — Ride Hailing

Mar 2018 → May 2019

RIYADH, KSA

+ Developed a ride-hailing ecosystem (iOS, watchOS) serving **15M+ users** in Saudi Arabia.
+ Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
+ Integrated Apple Pay, HyperPay, and push-notification infrastructure.
+ Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
+ Built the driver rating system and real-time fare calculation engine.

Senior iOS Engineer @ HungerStation — DeliveryHero

Sep 2016 → Mar 2018

KSA

+ Contributed to a major food-delivery platform serving **20M+ users**.
+ Optimised app performance for high-volume transaction processing.
+ Implemented real-time order tracking and delivery optimisation.
+ Collaborated with backend teams on API design and integration.

iOS Developer @ ALNapedh

May 2016 → Aug 2016

+ Developed food-ordering apps with Firebase integration and real-time features.
+ Implemented Spreedly payments and live-chat functionality.
+ Built responsive UI with Auto Layout across multiple device sizes.

iOS & Apple Watch Developer @ Nile Design Co.

Jul 2015 → Sep 2015

+ Early adopter of Apple Watch development (watchOS 1–2).
+ Built a foundation in iOS and watchOS app development.
+ Adopted SwiftUI from WWDC 2019 onward.

iOS Developer @ UK Pro. Solutions Ltd.

May 2014 → May 2015

+ Foundation iOS experience in Objective-C and early Swift.
+ Built social networks and e-commerce platforms.
+ Learned core iOS principles and App Store deployment.

## projects[]

selected

[01] ai / privacy

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur, live video monitoring.

◆ AI-powered privacy browser

[02] full-stack

Ride-Hailing Platform

SwiftUI · Kotlin · Jetpack Compose · NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js

End-to-end ride-hailing system built solo: a hardened NestJS backend (idempotency, double-entry ledger, 2,900+ tests) and a real-time Next.js ops console in production, with **four native clients** (Rider & Driver on iOS and on Android) in active development.

◆ Full-stack ownership · prod security

[03] delivery

HungerStation

UIKit · Firebase · Real-time APIs · Perf

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

[04] mobility

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · watchOS

Complete ride-hailing ecosystem: rider, driver, and Apple Watch apps. Real-time GPS and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

[05] enterprise

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 20k+ restaurants across Saudi Arabia. Offline-first architecture, complex table management, real-time orders.

★ 4.8 / 5 — 2,437 ratings

[06] community

Daleelo

MapKit · Core Location · Social SDKs

Muslim community app — prayer times, mosque finder, and community features with enhanced location + social integration.

★ 4.5 / 5 — 1,287 ratings

## stack.config

engineering_practice =
System Design · Automated Testing · API Design · Offline-first & Sync · Realtime (WebSockets) · Performance Profiling · CI/CD · Code Review · Modular Codebases · Versioned SDKs · Mentoring · Agile / Scrum

ios_2025 =
Swift 6.0 · SwiftUI · UIKit · Combine · async/await · SwiftData · iOS 18+

android_2025 =
Kotlin · Jetpack Compose · Android · Coroutines · Flow · MVVM

architecture =
MVVM · VIPER · Clean Architecture · Coordinator · TDD

apple_ecosystem =
watchOS · tvOS · macOS · WidgetKit · Live Activities · App Intents

intelligence =
Core ML · Vision · ARKit · Dynamic Island · App Tracking

backend_apis =
NestJS · TypeScript · PostgreSQL · TypeORM · Redis · JWT/OTP · REST · GraphQL · Firebase · CloudKit · WebSockets · Swagger

tooling =
Xcode 15+ · Fastlane · TestFlight · Instruments · Git · Copilot · Cursor

## portfolio/

FOOD & GAMING
- HungerStation
- Punters

TRANSPORTATION
- Ego — Rider
- Ego — Driver
- Ego — Watch

ENTERPRISE
- Marn POS
- Medical Guide

SOCIAL & COMMUNITY
- Daleelo
- NU-inc
- VeeU

WATCHOS
- Ego Watch
- TaQs Weather
- TweetMe

E-COMMERCE
- iBss
- Exclusive Luxe
- Scratch

RELIGIOUS & EDU
- Afasy
- ArabNileGroup

UTILITY & TOOLS
- Unlocking Smart
- Insta
- iContacts
- First iOS app

AI & ML
- HaramBlur
- RecognizeMe
- Grad project · OpenCV

WEB
- FaceDemo
- Web Pages

## record{}

// education

B.Sc. Information Systems
Computer Sciences College
Asyut University, Egypt · 2009-2013
iOS Semester — Excellent (91.5%)

// certification

ISTQB Foundation Level
Egyptian Software Testing Qualification Board

// languages

Arabic	native
English	professional

## honours[]

01 Owned the architecture of a POS platform serving **20k+ restaurants** and cafés for **7 consecutive years** — through its growth from launch scale to enterprise.

02 Held a **99.8-99.9% crash-free** rate and cut deployment time **60%** with the standards, tests and CI/CD the team ran on.

03 Architecting a **full-stack ride-hailing platform solo** — NestJS/PostgreSQL backend and Next.js ops console in production, four native clients (rider and driver, iOS and Android) in development.

04 Built a food-delivery platform serving **20M+ users** and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) & early SwiftUI adopter (WWDC 2019).

⎇ main | ● Senior Software Engineer · iOS · Android · Backend | Cairo → Riyadh | UTF-8 | TypeScript · Swift · Kotlin | ★ 12+ yrs hands-on